Worksheets for Math 252

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Preface

These exercises were assembled to accompany a Discrete Mathematics course at Calvin University.

Part I

Graphs

1 Introduction to Graphs

1.1 Graphs

A Graph G consists of two sets

- V, a set of vertices (also called nodes)
- E, a set of edges (each edge is associated with two[1] vertices)

In modeling situations, the edges are used to express some sort of relationship between vertices.

1.1.1 Graph flavors

There are several flavors of graphs depending on how edges are associated with pairs of vertices and what constraints are placed on the edge set. The first four are the most important.

graph type	edges	multi-edges allowed?	self-loops allowed?	cycles allowed?
simple graph	undirected	no	no	yes
tree	undirected	no	no	no
directed graph	directed	no	yes	yes
directed acyclic graph	directed	no	no	no
multigraph	undirected	yes	no	yes
directed multigraph	directed	yes	yes	yes
pseudo-graph	undirected	yes	yes	yes

multi-edges: multiple edges between the same two vertices (in the same direction, in the case of directed graphs). **self-loop**: an edge connecting a vertext to itself. **cycle**: a chain of edges (called a **path**) that returns to where it started.

Directed graphs are sometimes called **digraphs**. **Directed Acyclic Graphs** are sometimes called **DAGS**.

1.1.2 Graph Representation

(Small) graphs are often depicted using dots for vertices and line segments or arcs for edges. For directed graphs, an arrow is used to indicate the direction.

1. For each type of graph, draw a picture of an example graph with 5 vertices.
2. Why are trees called trees?

1.1.3 Labeled Graphs

In many situations it is useful to add additional information to either the vertices or the edges. These are sometimes called "labeled" graphs and the extra information referred to as labels.

[1]In some types of graphs, the two vertices may be the same, so there is really only one vertex. This is called a self-loop.

1.2 Graph Models

Many things can be models with graphs of various types. In each situation below,

- identify the vertices
- identify what edges represent
- determine which flavor of graph would be used
- determine what, if any, information might be useful as labels (on vertices or on edges or on both)
- if the graph has any special properties, note them
- if your group disagrees about some of the answers, identify whether it is because you are making different assumptions about the situation being modeled

Model	Vertices	Edges	Type of Graph	Labels/Properties?
3				
4				
5				
6				
7				
8				
9				
10				
11				

3. **Niche Overlap.** Some species of animals compete with other species in an ecosystem.
4. **Acquaintanceship.** Some pairs of people are acquainted with each other, some are not.
5. **Precedence and Concurrent Processing.** Computer programs can be executed more rapidly by executing some statements concurrently But it is important not to execute a statement that depends on the results of statements that have not yet been executed. What is the precedence graph for this simple program?

```
a = 0
b = 1
c = a + 1
d = b + a
e = d + 1
e = c + d
```

6. **Influence.** In studies of group behavior it is observed that certain people can influence the thinking of other people.
7. **Hollywood.** Some actors have been in movies together, others have not.
8. **Round-Robin Tournament.** In a round robin tournament, each team plays each other team once. (The graph should keep track of who wins and loses each game.)
9. **General Sports Schedule.** The schedule for most leagues is not a round-robin tournament. How does that change the graph? If the games haven't been played yet, what information might the graph keep track of? How?
10. **World Wide Web.**
11. **Road maps.**

1.2.1 Bonus Problem

From 1961 through 1977 the NFL (National Football League) had a 14-game season. Until 1976, when two new teams were added, there were 13 teams in each of 2 conferences. The NFL wanted a schedule were each team would play 11 games against other teams in their conference and 3 games against teams from the other conference.

Devise such a schedule or explain why it is not possible. For simplicity, let's name the teams in the American Football Conference (AFC) $A_1, A_2, \ldots, A_{13}$ and the teams in the National Football Conference (NFC) $N_1, N_2, \ldots, N_{13}$.

2 More Graphs

2.1 Some Example Graphs

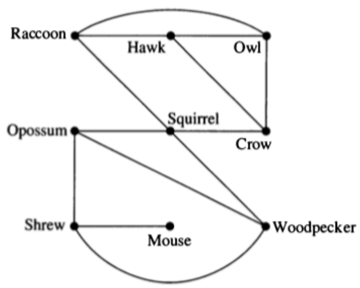


FIGURE 6 A Niche Overlap Graph.

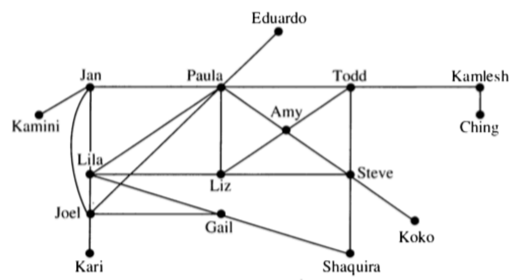


FIGURE 7 An Acquaintanceship Graph.

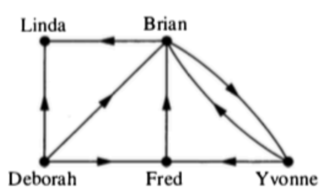


FIGURE 8 An Influence Graph.

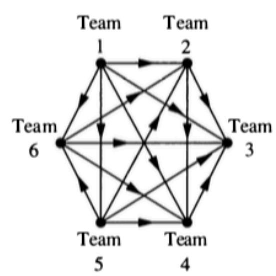


FIGURE 9 A Graph Model of a Round-Robin Tournament.

S_1 $a := 0$
S_2 $b := 1$
S_3 $c := a + 1$
S_4 $d := b + a$
S_5 $e := d + 1$
S_6 $e := c + d$

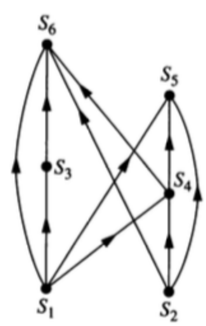


FIGURE 11 A Precedence Graph.

2.2 Some Terminology

- Two vertices are **adjacent** if they are connected by an edge.
- Two edges are **incident** if they share a vertex.
  - For directed graphs, one edge must point into the vertex and one out.
- The **degree** of a vertex in an undirected graph is the number of edges that include the vertex;
  - Self-loops (if they are allowed) contribute 2 to the degree.
  - For a directed graph, we can talk about **in-degree**, **out-degree**, and **total degree**.
- A **path** in a graph is a sequence of edges, each one incident to the next.
  - If multiple edges are not allowed, this can also be described as a sequence of vertices, each one adjacent to the next.
  - For directed graphs, we require that the directions of the edges be compatible.
- A **cycle** is a path that begins and ends at the same vertex.
- A **connected graph** is one where there is a path between any pair of vertices.
  - For a directed graph, we *ignore* the direction of the edges.
- H is a **subgraph** of G if the vertex set of H is a subset of the vertex set of G and the edge set of H is a subset of the edge set of G, and H is a graph.

2.3 Some Questions

These questions will help make sure you understand the terminology above.

1. In figure 6, which species compete with squirrels?
2. Make a table showing the degree of each vertex in Figure 7. What does a large degree indicate about a person? A small degree?
3. Consider the graph in Figure 6.
   a. Compute the degree of each vertex in Figure 6.
   b. Now add up all the degrees.
   c. How many edges are in the graph?
   d. How is the sum of all the degrees related to the number of edges in the graph? Why? Does this hold for all graphs?
   e. This result is called the **Handshaking Theorem**. Why does it have that name?
4. In Figure 9, assume that $a \to b$ means team a defeated team b.
   a. Compute the in-degree and out-degree of each team in Figure 9.
   b. What do these numbers tell us about the teams?
   c. Who is the winner of the Round-Robin tournament in Figure 9?

5. Consider Figure 8.
    a. Compute the in-degree and the out-degree of each vertex in Figure 8.
    b. Add up all the in-degrees.
    c. Now add up all the out-degrees.
    d. What do you notice? Will this hold for all directed graphs, or is this graph special?
6. What is the longest path you can find in Figure 8? What does it represent in terms of the model?

2.4 NFL

From 1961 through 1977 the NFL (National Football League) had a 14-game season. Until 1976, when two new teams were added, there were 13 teams in each of 2 conferences. The NFL wanted a schedule were each team would play 11 games against other teams in their conference and 3 games against teams from the other conference.

Suppose we create such a schedule for the NFL. Consider the part of the schedule that includes only the 13 NFC teams. We can represent this as a directed graph (road team $\rightarrow$ home team). (This graph would be a subgraph of the graph for the entire schedule.)

7. How many vertices does this graph have?
8. What is the total degree of each vertex? Why?
9. What is the sum of all the total degrees?
10. How many edges does this graph have? Why is this impossible?

So a little bit of graph theory shows that the NFL's desired schedule is not possible.

2.5 Some special graphs

- K_n, the complete graph with n vertices.
    - simple graph with every possible edge.
- C_n, the cycle with n vertices.
    - simple graph that consists of a single cycle connecting all the vertices and no other edges.
- W_n, the wheel with $n+1$ vertices.
    - Formed by adding 1 vertex to C_n and an edge between the new vertex and all the vertices in C_n.
- Q_n, the n-dimensional cube.
    - vertices: bit strings with n bits
    - edges: two vertices are adjacent if and only if their bit strings differ in exactly one position.

11. Draw K_5. How many edges does K_5 have? How many edges does K_n have?
12. Draw C_5. How many edges does C_5 have? How many edges does C_n have?
13. Draw W_5. How many edges does W_5 have? How many edges does W_n have?
14. How many vertices does Q_3 have? How many edges does Q_3 have? Draw Q_3.
15. How many vertices does Q_4 have? How many edges does Q_4 have?
16. In Figure 7 there is a subgraph that is a K_4. List its vertices. Is there a subgraph that is a K_5? How do you know?

3 Graph Representations

3.1 4 Representations of Graphs

There are several ways to represent a (simple) graph. The efficiency of some graph algorithms depends on which representation is used to store the graph.

3.1.1 Edge Lists

An edge list simply lists off the edges (i.e., pairs of vertices) in the graph.

3.1.2 Adjacency Lists

For each vertex i, a list of vertices j such that there is an edge from i to j.

3.1.3 Adjacency Matrix

An adjacency matrix A is a square matrix with a row (and column) for each vertex.

- $A_{ij} = 1$ if there is an edge from vertex i to vertex j
- $A_{ij} = 0$ if there is **not** an edge from vertex i to vertex j

3.1.4 Incidence Matrix

An incidence matrix M has a row for each vertex and a column for each edge.

- $M_{ij} = 1$ if edge j is incident with vertex i.
- $M_{ij} = 0$ if edge j is **not** incident with vertex i.

3.2 Representation to Graphs

Draw graphs with the following representations.

1. adjacency list

vertex	adjacent vertices
1	2, 3, 5
2	1
3	1, 4, 5
4	3, 5
5	1, 3, 4

2. edge list: $\{\{1,2\},\{2,3\},\{3,4\},\{4,5\},\{1,3\},\{2,4\},\{3,5\}\}$.

3. adjacency matrix:

```
0 1 1 1
1 0 1 1
1 1 0 0
1 1 0 0
```

4. adjacency matrix:

```
0 1 1 0
1 0 0 1
1 0 0 1
0 1 1 0
```

5. adjacency matrix

```
0 1 0 0 1 1
1 0 1 1 0 1
0 1 0 0 1 1
0 1 0 0 1 0
1 0 1 1 0 1
1 1 1 0 1 0
```

6. incedence matrix

```
1 1 0 0 0 0
0 0 1 1 0 1
0 0 0 0 1 1
1 0 1 0 0 0
0 1 0 1 1 0
```

7. incedence matrix

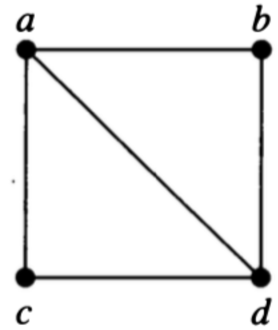
```
1 1 1 0 0 0 0 0
0 1 1 1 0 1 1 0
0 0 0 1 1 0 0 1
0 0 0 0 0 0 1 1
1 0 0 0 1 1 0 0
```

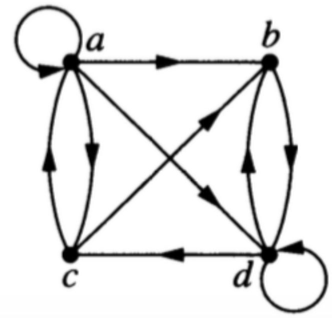
Variations on the theme

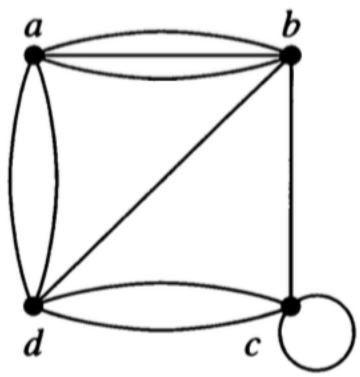
8. Which of these representations can be used (perhaps with slight modification) with **directed graphs**? What adjustments do you need to make?
9. Which of these representations can be used (perhaps with slight modification) for graphs with **multi-edges**? What adjustments do you need to make?
10. Which of these representations can be used (perhaps with slight modification) for graphs with **self-loops**? What adjustments do you need to make?

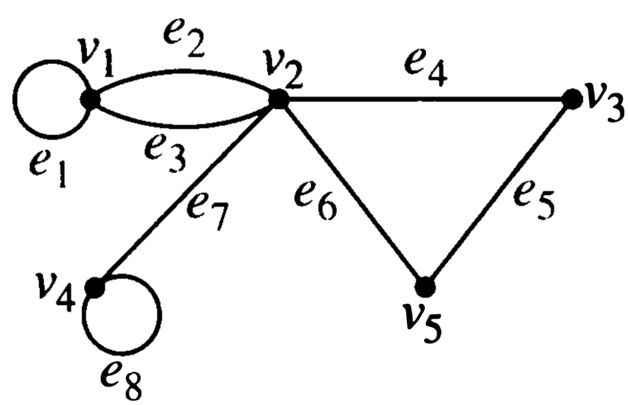
3.3 Graph to representation

11. Use each representation to represent the following graphs or say why it isn't possible.









4 Graph Isomorphism

Two (mathematical) objects are called **isomorphic** if they are "essentially the same" (iso-morph means same-form). What "essentially the same" means depends on the kind of object. For graphs, we mean that the vertex and edge structure is the same. So

> Graphs G and H are **isomorphic** if there is a bijection (1-1 and onto function)
>
> $$f : G \to H$$
>
> such that there is an edge from a to b in G if and only if there is an edge from $f(a)$ to $f(b)$ in H.
> The function f is called an **isomorphism**.

(For graphs that allow multi-edges, we require that the number of edges be the same.)

More informally, **to show that two graphs are isomorphic, we must show how to match up vertices in one graph with vertices in the other so that edges are preserved.**

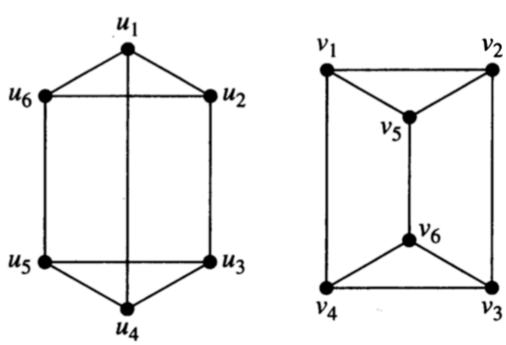
4.1 Showing graphs are isomorphic

When graphs are isomorphic, we can demonstrate this by providing the isomorphism.

1. Show that the following pair of graphs is isomorphic by providing the isomorphism.

   a. What must you demonstrate to convince someone that your function is indeed an isomorphism?

   b. Is there more than one isomorphism? If so, give another one.



4.2 Showing graphs are not isomorphic

Showing that two graphs are *not* isomorphic can be trickier. Somehow you need to demonstrate that no isomorphism exists. Unfortunately, there are a lot of bijections ($n!$ if the graphs each have n vertices), so checking all possible bijections to see if they are isomorphisms would not be a very efficient algorithm.

News Flash: Until recently, the best known algorithm for Graph Isomorphism had exponential worst case running time (but was polynomial time on average). In 2015, Laszlo Babai demonstrated an algorithm that runs in sub-exponential time (almost, but not quite polynomial time).

Back to showing two graphs are not isomorphic. We need to demonstrate that no isomorphism exists, but we don't want to check all of them. One good approach is to consider **graph invariants**. A graph invariant is a property that is the same whenever two graphs are isomorphic.

2. Which of the following are graph invariants?

    a. number of vertices
    b. number of edges
    c. degree sequence
    d. adjacency matrix
    e. incidence matrix
    f. having a subgraph isomorphic to K_4
    g. having a subgraph isomorphic to ________ (can you fill in the blank with any graph?)

Unfortunately, matching invariants does not guarantee that graphs are isomorphic, but you can use graph invariants to help narrow the search.

3. Explain why each of the following pairs of graphs is not isomorphic.

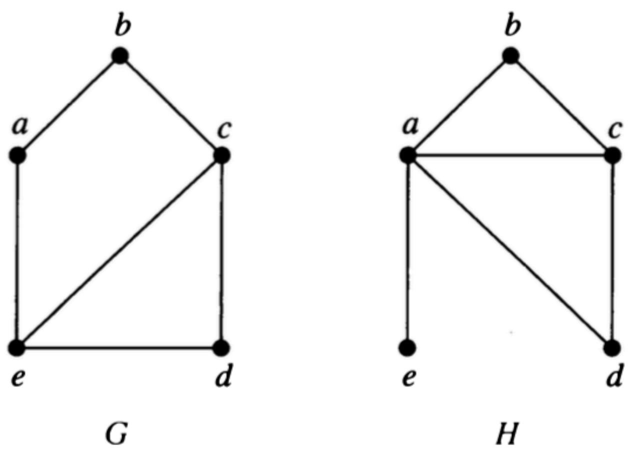


Figure 4.1: a

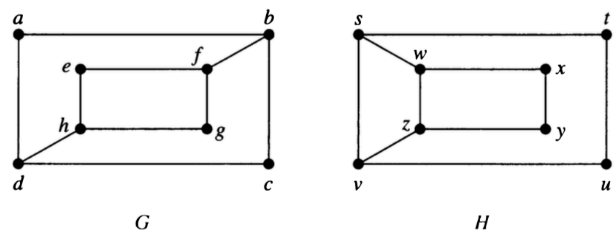


Figure 4.2: b

4.3 Isomorphic or not?

4. For each pair, determine whether the graphs are isomorphic. Be sure to explain how you know.

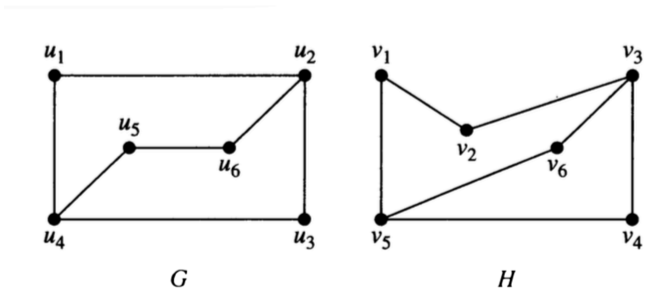


Figure 4.3: a

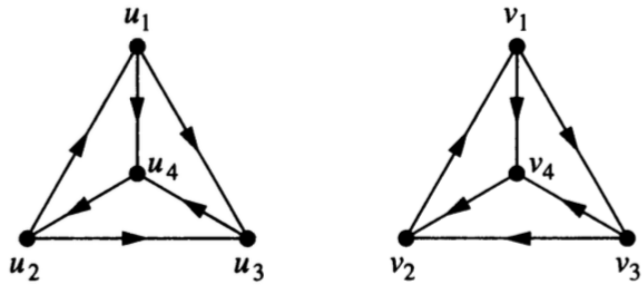


Figure 4.4: b

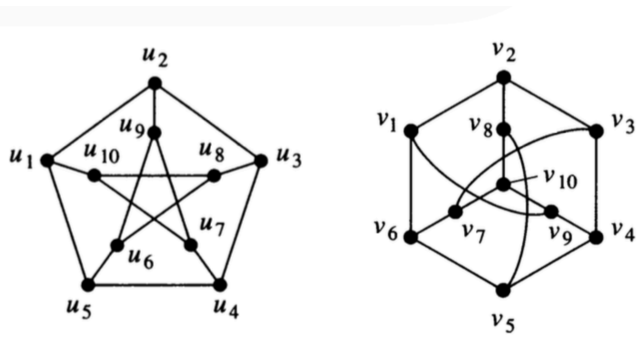


Figure 4.5: c

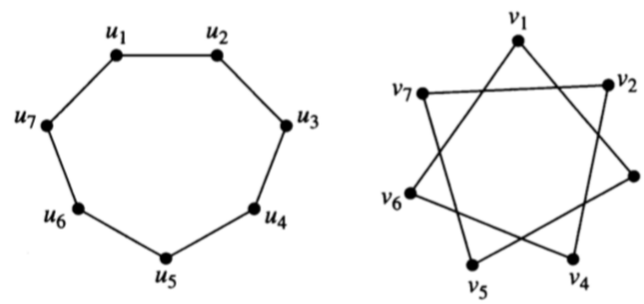


Figure 4.6: d

5 Paths in Graphs

5.1 Definition of a path

- Informally, a **path** in a graph is a sequence of edges, each one incident to the next.
  - Can sometimes be described as a sequence of vertices, each one adjacent to the next (see below).
  - For directed graphs, we require that the directions of the edges be compatible.
- More formally, let n be a nonnegative integer and G an undirected [directed] graph. A **path** of **length** n from vertex v_0 to vertex v_n in G is a sequence of n edges $e_1, \dots, e_n$ of G such that when $1 \le i \le n$, then $e_i = \{v_{i-1}, v_i\}$ $[e_i = \langle v_{i-1}, v_i \rangle]$
- A path is a **circuit** (also called a **cycle**) if it begins and ends at the same vertex and has length ≥ 1.
- A path or circuit is **simple** if it does not include the same edge more than once.

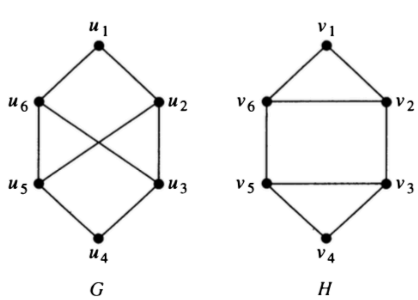
Questions

1. What is a path of length 0?
2. Can a path of length 1 be a circuit? If so, draw an example. If not, explain why not.
3. Why are paths defined in terms of edges rather than vertices? In what situations does it matter? When does it not matter?

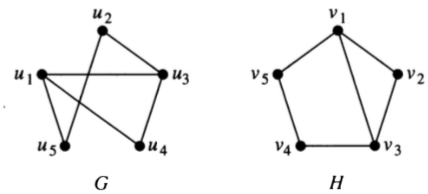
5.2 Paths and Isomorphism

Paths can be useful for finding an isomorphism between G and H or for showing that there is no isomorphism.

4. Consider the pair of graphs below.
   a. Compute the degreee sequences for each graph.
   b. Explain why any isophormphism must either pair u_1 with v_1 and u_4 with v_4 or else pair u_1 with v_4 and u_4 with v_1.
   c. How long is the shortest path from u_1 to u_4? the shortest path from v_1 to v_4?
   d. Is there a path of length 4 between u_1 and u_4? between v_1 and v_4?
   e. Are the graphs isomorphic? Explain.



5. How about these graphs? Are they isomporphic? How can considering paths help you figure this out?



5.3 Euler and Hamilton Paths

- An **Euler path** is a path that visits every edge of a graph exactly once.
- A **Hamilton path** is a path that visits every vertex exactly once.

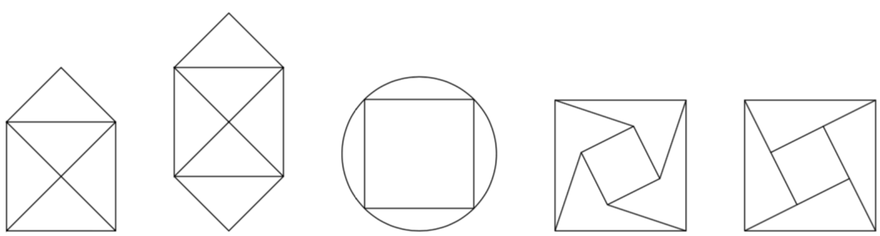
Euler paths are named after Leonid Euler who posed a famous problem about the bridges in Königsberg. Euler is pronounced “Oiler”.

Hamilton paths are named after the Irish mathematician William Rowan Hamilton, not after the American politician Alexander Hamilton.

5.3.1 Can you trace it?

Perhaps you have seen the following type of “puzzle”. The goal is to trace the figures without lifting your pencil from the paper and without retracing any of the lines.

6. Which kind of path is this puzzle asking for?
7. Which of these can you trace in this manner?



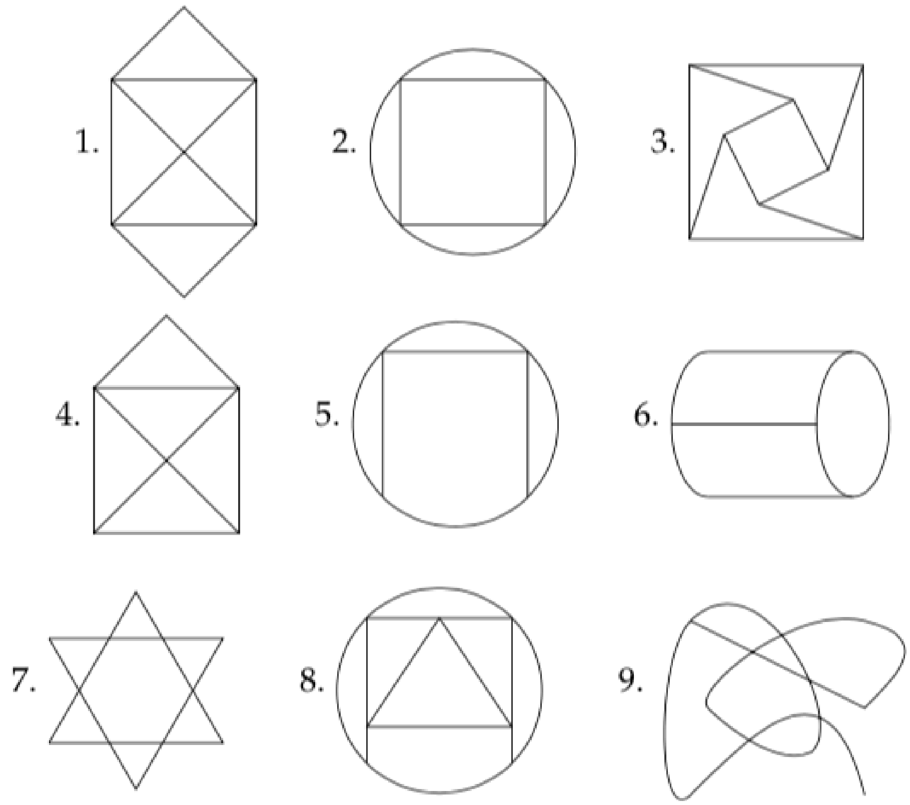
5.4 Relationships in Graphs

A **planar graph** is a graph that *can be* drawn in the plane without any edges crossing.

8. Show that K_4 is a planar graph by drawing it without any edge crossings.

9. Show that Q_3 is a planar graphs by drawing it without any edge crossings.

10. Here are some graphs that are missing their vertices. Add dots showing vertices so that each graph is a planar graph.



11. Fill in the table below for the graphs above. Do you notice any patterns? Can you make any conjectures? Can you prove any of the conjectures? (Note: an even vertex is a vertex with even degree. An odd vertex is a vertex with odd degree.)

Graph	number of even verticies	number of odd vertices	total degree	number of vertices	number of edges	number of regions	Euler path?	Euler circuit?
1								
2								
3								
4								
5								
6								
7								
8								
9								

5.5 Connectedness

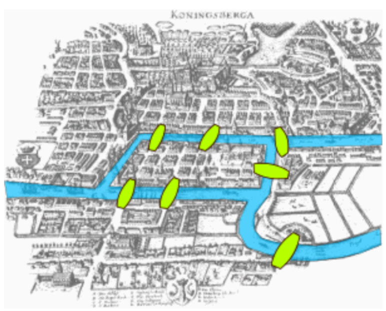
- An **undirected graph** is **connected** if there is a path between every pair of vertices.
- A **directed graph** is **strongly connected** if there is a path between every pair of vertices.
- A **directed graph** is **weakly connected** if the underlying undirected graph is connected.
- A connected component H of a graph G is maximal connected subgraph. That is,
  - H must be a subgraph of G,
  - H must be connected, but
  - H cannot be a subgraph of a larger connected subgraph of G.

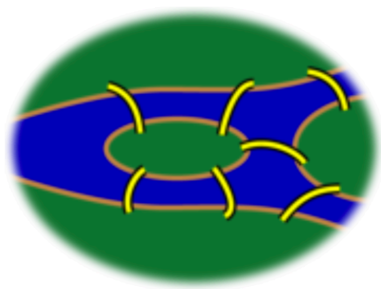
12. Draw a directed graph with 6 vertices that is weakly connected but not strongly connected.
13. Draw an undirected graph with two connected components. Count the number of vertices, edges, and regions in your graph. How does this compare to the results in the table in problem 11?

5.6 Königsberg Bridge Problem

5.6.1 Original Version

One of the founders of graph theory was Leonid Euler (one of the greatest mathematicians who ever lived). At the time he was alive, the city of Königsberg had seven bridges connecting the two banks of the river flowing through downtown and two islands in that river. Here are a couple maps of downtown Königsberg at the time. One is a bit stylized.



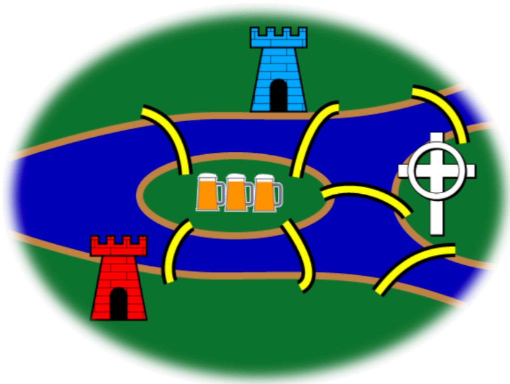


As the story goes, the residents of Königsberg enjoyed walking the bridges. The question is this: How can someone take a Sunday afternoon walk that starts and ends in the same place and crosses each bridge exactly once? (Leaving the map, renting hot air balloons, swimming, etc. are not allowed.)

We'll return to this problem in just a moment. But first, a different sort of puzzle.

1. Create a graph that represents the Königsberg Bridge Puzzle (so it can be traced in this manner, if and only if the Königsberg Bridge Puzzle has a solution). What are the vertices? What are the edges? What type of graph is it?

2. Can you trace the Königsberg Bridge graph?



5.6.2 Extentions to the Königsberg bridge problem

Here are some extensions to the Königsberg Bridge problem. First, we embelish the map a bit: The northern bank of the river is occupied by the Schloss, or castle, of the Blue Prince; the southern by that of the Red Prince. The east bank is home to the Bishop's Kirche, or church; and on the small island in the center is a Gasthaus, or inn.

It being customary among the townsmen, after some hours in the Gasthaus, to attempt to walk the bridges, many have returned for more refreshment claiming to have walked each bridge exactly once. However, none have been able to repeat the feat by the light of day.

3. **The Blue Prince's eighth bridge.** The Blue Prince, having analyzed the town's bridge system by means of graph theory, concludes that the bridges cannot be walked. He contrives a stealthy plan to build an eighth bridge so that he can begin in the evening at his castle, walk the bridges, and end at the Gasthaus to brag of his victory. Of course, he wants the Red Prince to be unable to duplicate the feat. Where does the Blue Prince build the eighth bridge?

4. **The Red Prince's ninth bridge.** The Red Prince, infuriated by his brother's solution to the problem, wants to build a ninth bridge, enabling him to begin at his castle, walk the bridges, and end at the Gasthaus to rub dirt in his brother's face. Furthermore, his brother should then no longer walk the bridges himself. Where does the Red Prince build the ninth bridge?

5. **The Bishop's tength bridge.** The Bishop has watched this furious bridge-building with dismay. It upsets the townspeople and, worse, contributes to excessive drunkenness. He wants to build a tenth bridge that allows all the inhabitants to walk the bridges and return to their own beds. Where does the Bishop build the tenth bridge?

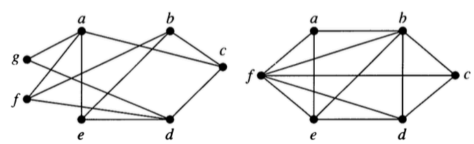
6 Bipartite Graphs

6.1 Definition

A simple graph G is **biparatite** if its vertices can be partitioned into two disjoint subsets V_1 and V_2 and every edge connects a vertex in V_1 with a vertex in V_2. The pair $\langle V_1, V_2 \rangle$ is called a **bipartition** of G.

6.2 Examples

1. Which of the following are bipartite?



G H

2. For what values of n are the following bipartite?

   a. K_n
   b. C_n
   c. Q_n

3. Can a bipartite graph have more than one bipartition? If so, give an example. If not, explain why not.

4. Describe an algorithm for checking whether a graph is bipartite. How efficient is your algorithm?

6.3 Complete Bipartite Graphs

A complete biparite graph $K_{m,n}$ has two disjoint sets of vertices V_1 and V_2 with m and n elements. There is an edge between x and y if and only if $x \in V_1$ and $y \in V_2$ or $x \in V_2$ and $y \in V_1$.

Here are some examples.

$K_{2,3}$ $K_{3,3}$

$K_{3,5}$ $K_{2,6}$

5. Draw the following graphs: $K_{2,2}$, $K_{4,2}$, $K_{4,3}$

7 Planar and Non-planar Graphs

A graph is a **planar graph** if it can be drawn in the plane with no edge crossings. [Note: You may need to rearrange the vertices and edges to avoid crossings.]

7.1 Euler and Planarity

We have already seen **Euler's Formula for Planar Graphs**:

$$v - e + r = 1 + \text{number of connected components} \ ,$$

where v is the number of vertices, e the number of edges and r the number of regions.

In particular, for a connected graph we have $v - e + r = 2$.

1. Show that the following are planar.

   a. $K_{2,2}$
   b. $K_{2,3}$
   c. $K_{2,4}$

2. Is $K_{2,n}$ planar for every n?
3. Use Euler's formula to show that K_5 is not planar.

   a. How many vertices does K_5 have?
   b. How many edges does K_5 have?
   c. If K_5 were planar, how many regions would it have?
   d. Show that K_5 can't have that number of regions, so it must not be planar.

4. Use Euler's formula to show that $K_{3,3}$ is not planar.

7.2 Kuratowski's Theorem

The proof of the following theorem is too involved for this course, but it turns out that all nonplanar graphs contain a "homeomorphic copy" of either K_5 or $K_{3,3}$.

Kuratowski's Theoerm: A graph G is nonplanar if and only if it contains a subgraph H that is homeomorphic to either K_5 or $K_{3,3}$.

- an **elementary subdivision** of a graph G removes one edge (u, v) and adds a new vertex n and two new edges (u, n) and (v, n).
- a **subdivision** of a graph G is the result of performing 0 or more elementary subdivisions, starting from G.
- two graphs are **homeomorphic** if some subdivision of one is isomorphic to some subdivision of the other.

5. Create a graph with 5 vertices and 7 edges. Now perform three elementary subdivisions on your graph.
6. Describe what an elementary subdivision is in simple words a kindergarterner could understand.
7. Explain why two homeomorphic graphs are either both planar or both nonplanar.
8. Use Kuratowski's Theorem to show that the Petersen graph is nonplanar.

   There is more than one way to do this, but if you are having trouble, here are some things you could try.

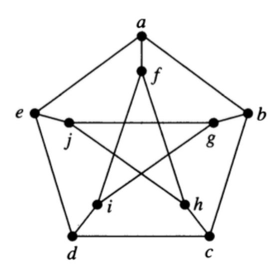
   a. Frist, decide whether you are looking for a version of K_5 or a version of of $K_3, 3$ inside the Petersen graph. (It's pretty easy to give an argument that one of these two will not work. Identifying which graph to work with will save you going down a wrong path right from the start.)

   b. Now see if you can identify

      - the 5 or 6 vertices in the Petersen graph that will corresponsd to the vertices in K_5 or $K_{3,3}$.
      - any vertices or edges that you will remove from the Petersen graph. (We only need to work with a subgraph, so we are allowed to remove things if they get in the way.)
      - any vertices in the Petersen graph that will remain, but not be one of the vertides corresponding to K_5 or $K_{3,3}$. These must be the vertices invovled in subdivisions.

      There may be some trial and error involved here.

   c. When you think have identified everything correctly, double check to make sure the you have the correct structure for K_5 or $K_{3,3}$.



9. Show that the Petersen graph is not planar *without* using Kuratowski's Theorem.

Part II

Counting and Probability

8 Some Counting Problems

Each of these problems involves determining how many of something there are. For most, there will be too many to simply make a list and count them off, so you will need to "count without counting". As you go along, make a list of important counting principles you are using.

1. There are 32 microcomputers in a computer center. Each microcomputer has 24 ports. How many different ports to a microcomputer in the center are there?
2. A new company with just two employees, Sanchez and Patel, rents a floor of a building with 12 offices. How many ways are there to assign different offices to these two employees?
3. The chairs of an auditorium are labeled with a letter (indicating the row) and a positive integer not exceeding 100 (indicating the seat within the row). Give an upper bound on the number of seats in the auditorium.
4. How many bitstrings of length 8 are there?
5. How many bitstrings of length 8 either start with a 1 or end with a 1?
6. A state issues license plates that consist of three letters followed by three numbers. How many such plates are there?
7. A graph isomorphism is a 1-1 and onto function from the vertices in one graph to the vertices in another (that additionally preserves edges). You decide to write a little program that will check each 1-1 and onto function to see if it an isomorphism.

   If your graphs have n vertices, how many such functions must you check?
8. Now you modify your graph isomorphism algorithm so that it only tries bijections (1-1 and onto functions) that respect the degrees of the vertices. If the degree sequences of two graphs are both $(5, 5, 4, 4, 4, 3, 3, 2)$, how many functions must you check? How much better is this than your previous algorithm (on these two graphs)?
9. Let's use the following coding scheme to describe possible telephone numbers.

   - let X denote a digit that can take on any integer value $0 - 9$;
   - let N denote a digit that can take on values $2 - 9$
   - let Y denote a digit that can only take on values 0 or 1.

   a. In the scheme used in the 1960's, telephone numbers had to have the form NYX-NNX-XXXX. How many such phone numbers are there?

   b. A new scheme (*The North American Numbering Plan*) was introduced that required telephone numbers have the form NXX-NXX-XXXX. How many more phone numbers are available in the new scheme?
10. How many password of length 6, 7, or 8 are there if the passwords must be alpha-numeric, capitalization doesn't matter, and there must be at least one digit?
11. In the previous example, how many passwords are there if the first character must be a letter?
12. Professor Pruim puts the names of 30 students into a hat. He draws one out and gives that person a chocolate bar. Then he draws out another name, and gives that person a bag of M and M's. Finally, he draws out a third name and gives that person a bag of Valentine hearts he picked up cheap on February 15.

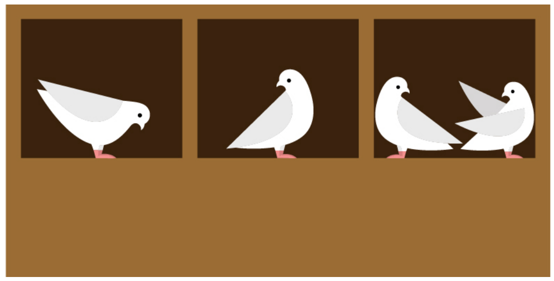
    a. How many different ways can the prizes be distributed?

    b. If you are one of the 30 students, how many of those ways include you getting a prize?

Answers/Solutions

1. `32 * 24` = 768.
2. `12 * 11` = 132.
3. `26 * 100 = 2600`
4. $2^8 = 256$
5. Three ways:
    a. only bad thing is beginning AND ending with 0, so $2^8 - 2^6 = 192$.
    b. Start with 1 or end with 1 but not both: $2^7 + 2^7 - 2^6 = 192$
    c. Three ways to start/end (0-1, 1-0, or 1-1), and 2^6 ways to fill in the middle six bits: $3 \cdot 2^6 = 192$.
6. $10^3 \cdot 26^3 = 17576000$
7. $n!$
8. $2 \cdot 1 \cdot 3 \cdot 2 \cdot 1 \cdot 2 \cdot 1 \cdot 1 = 24$. ($8! = 40320$, so that's a pretty big speed up.)
9. See book.
10. $36^6 - 26^6 + 36^7 - 26^7 + 36^8 - 26^8 = 2684483063360$
11. `26 * (36^5 - 26^5) + 26 * (36^6 - 26^6) + 26 * (36^7 - 26^7)` = 1878468937280
12. a. `30 * 29 * 28` = 24360
    b. `1 * 29 * 28 + 29 * 1 * 28 + 29 * 28 * 1` = `30 * 29 * 28 - 29 * 28 * 27` = 2436

9 Pigeonhole Principle



Big Idea: It is not possible for everyone to be above average or for everyone to be below average. (It is possible for everyone to be exactly average.)

So if n things are distributed into k groups,

- The largest group will have at least $\lceil n/k \rceil$ things, and
- The smallest group will have at most $\lfloor n/k \rfloor$ things.

This is called the (generalized) pigeon hole principle based on the analogy of putting letters into mail slots (called pigeon holes).

Applying the pigeonhole principle is generally easy *once you figure out what the pigeons (letters, really) and the pigeonholes are*. Figuring out the pigeons and holes can be easy, or it can take a clever idea. (Some famous proofs center on very clever uses of the pigeonhole principle.)

1. Suppose (in some alternative reality) that Professor Pruim brought 150 cookies to a class with 34 students. At the end of the hour, all of the cookies had been eaten by the students.

   a. What is the average number of cookies eaten by a student?
   b. This means that the student who had the most cookies had at least ___________ cookies.
   c. This means that the student who had the fewest cookies had at most ___________ cookies.

In each of the following problems, each time you use the pigeonhole principle, identify the "pigeons" and the "holes" and explain how that solves the problem. If you don't use the pigeonhole principle directly, be sure to explain your reasoning carefully.

2. Fill in the blanks.

   a. In a class of 38 students, there are always at least ________ students who were born in the same month.

   b. In a class of 38 students, there are always at least ________ students who were born on the same day of the week.

3. In the 2014 Winter Olympics, 88 nations competed in 98 events. Explain how this guaranteed that some country would win more than one gold medal. (No, the answer is not "Norway was there.")

4. At the 2014 Winter Olympics, Norway won 11 gold medals. Does that fact alone guarantee that a country won no gold medals?

5. Joe has 21 white socks and 15 black socks. (Don't ask what happened to the missing socks, no one knows where they go.)

   a. If he pulls socks in the dark (so he can't tell what color they are), how many must he grab to make sure he has a pair of white socks?

   b. If he pulls socks in the dark (so he can't tell what color they are), how many must he grab to make sure he has a pair of black socks?

   c. If he pulls socks in the dark (so he can't tell what color they are), how many must he grab to make sure he has a pair of matching socks?

6. For his birthday, Joe received a gift of 8 red socks (4 pairs, and he hasn't lost any yet). Now how many socks must Joe grab to be sure that he has a matching pair?

7. Joe is going on a trip and needs 3 pairs of matching socks. He's still too lazy to turn on the light to see what socks he is grabbing. How many socks must Joe bring along to make sure he has 3 matching pairs? (He doesn't care how many pairs are red, white, or black.)

8. Checkers.

   a. Consider a 3×9 grid (like an odd-shaped checker board). Show that no matter how you place red and black checkers on the 27 squares, there will always be a rectangle whose 4 corners have the same color checker. (The rectangles must be at least 2×2; 1×1 rectangles don't have four different checkers on their corners.)

   b. Consider a 3×7 grid (like an odd-shaped checker board). Show that no matter how you place red and black checkers on the 21 squares, there will always be a rectangle whose 4 corners have the same color checker. (The rectangles must be at least 2×2; 1×1 rectangles don't have four different checkers on their corners.)

   c. Show how to place red and black checkers on a 3×6 grid without having any rectangles with mono-chromatic corners.

9. A computer lab has 15 workstations and 10 servers. Workstations can only access servers via direct connections, and each server can only use one of these connections at a time.

   a. If we connect every workstation to every server, how many connections is that?

   b. Show that 60 connections suffice make it possible to use any set of 10 workstations simultaneously. (No matter what we do, we won't be able to connect 11 workstations to servers since there are only 10 servers.)

   c. Show that 59 connections is not enough.
      That is, no matter how you do the connecting, there will always be a combination of 10 workstations that cannot simultaneously connect to servers. (Hint: on average, how many connections are there per server?)

10. Show that there is an integer that is divisible by 35 and consists only of 1's and 0's in its decimal representation.

    Hint: Consider the numbers 1, 11, 111, 1111, 11111, etc. Show that two of them have the same remainder when you divide by 35. How does that help you answer the question?

11. Was there anything special about the number 35 in the previous example? For what other divisors does this work? Explain.

12. Is there anything special about the digits 1 and 0 in the previous example? What other combinations of digits would work?

13. The integers from 1 to 10 are distributed around a circle. Prove that there must be three neighbors whose sum is at least 17. What about 18? What about 19?

    Hint: think about the nine numbers that are greater than 1.